

Meghan Tennant- Materials Engineering Reflection

In this module, we made a sensor utilizing an input of water vapor and an output of a change in resistance. To accomplish this, we first learned about what it is that material science engineers do, which is to find new ways to modify the structure and composition of materials to modify their function, whether it;s electronically, chemically, optically, or mechanically. We also learned what a sensor is, which is a device that takes input and results in output. Our sensor had an input of water vapor, so we first tried making one with just water, graphite, paper, and our multimeter. We used graphite as a conductor and the paper as the insulator, then tried using the multimeter to find variations in resistance.

The second day of this module, we toured Wilbur Powerhouse, obtaining stencils for IEDs, or Interdigitated Electrodes, which were cut out using the laser cutter. We also determined which materials would be best for constructing our sensors. For substrates, we found that paper, glass, silica, quartz, and polymers such as plastic and rubber would work well as insulators to build our electrodes on. For our conductors, we found that metals like copper, electrolytes like gatorade and mineral water, and carbon, specifically graphite, would work well, allowing electrons to move. In the lab, we started building the base for our sensor, layering different pieces of tape on glass slides parallel to one another to use as a place to deposit the conducting materials. We also placed our IED stencil on one of the glass slides to use as another option.

On the next day of the module, we toured the Sinclair Cleanroom, learning about MOCVD, which stands for metal organic chemical vapor deposition, and photolithography. We also learned what you could do in each of these rooms, and all of the applications that exist for these technologies. In the lab, we continued building our sensors, depositing the conductive metallic materials in our stencils. The materials we used were silver paste, silver paint, and carbon paint. The silver paint worked the best as it was not as chunky as the paste and didn't flake off as easily as the carbon paint. We used both glass slides and pieces of rubber to smooth out the metallic materials, with the glass slides working much better, getting a smoother layer of paint/paste. After the paint/paste dried, we peeled off the tape stencils and experimented with putting some on the hot plate to dry completely while letting others air dry. We did not notice a

difference in the process of drying. We then measured the conductivity of each material using the multimeter, with carbon paint having the highest by far, followed by the silver paint.

On the last day of this module, we built paper circuits shaped like butterflies, drawing the circuit schematic and learning about polarized vs non-polarized components. We also learned the key ideas to having a complete circuit. Then, we finished building our sensors, using the process of spincoating to spread an extremely thin layer of PEDOT-PSS, allowing the sensor to detect a change in conductivity/resistivity due to the double bonds of carbon in PEDOT-PSS that break connection when they come in contact with water. After letting our sensors completely dry, we tested the resistivity of each one before and after breathing on it. We determined that after we breathed on it, depositing water vapor into the sensor, the resistivity increased. Our sensor with the silver paint had nearly a 66% increase in resistivity.

Overall, while I found this module to be very confusing in the beginning, I started to understand it more as it went on and we continued building our sensors. I enjoyed getting to tour so many different parts of campus and see so many different, new machines that I'd never seen before nor heard of. My group also worked together well, each taking on an integral role in the process of making our sensor. I definitely learned a lot in this module, both from the professors and my classmates around me. My favorite things I learned were photolithography and the spincoating with PEDOT-PSS. I feel very accomplished as starting this module, I knew absolutely nothing about any of the things we did.